MC311 Requests Analysis

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Datasets and Ingestion

MC311 Requests

- Variables such as opened/closed date, area of service, zip code
- Source: dataMontgomery

Maryland Weather Summaries

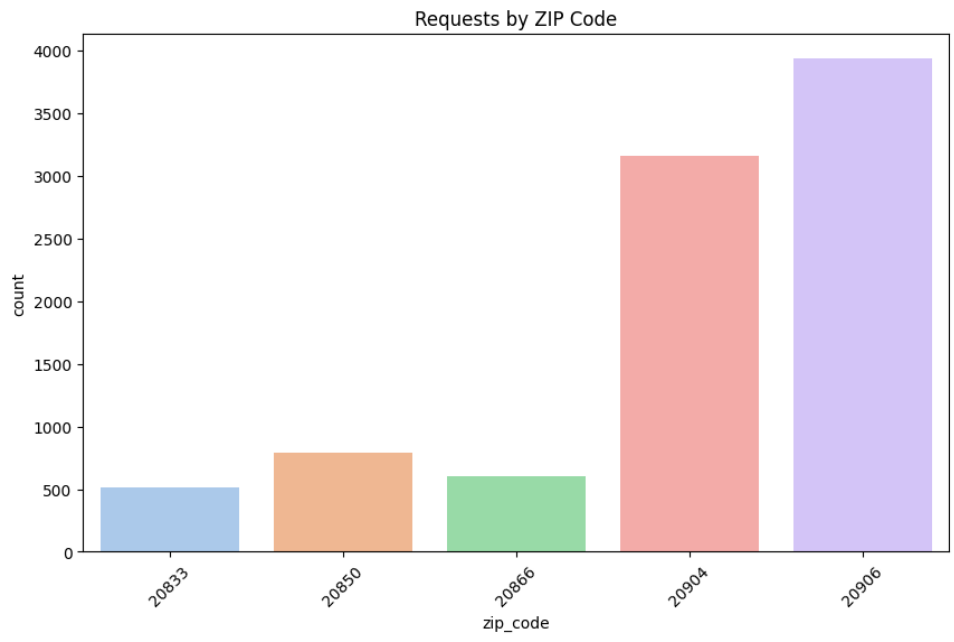
- Variables such as date recorded, max/min temperature, precipitation
- Source: National Oceanic and Atmospheric Administration (NOAA)

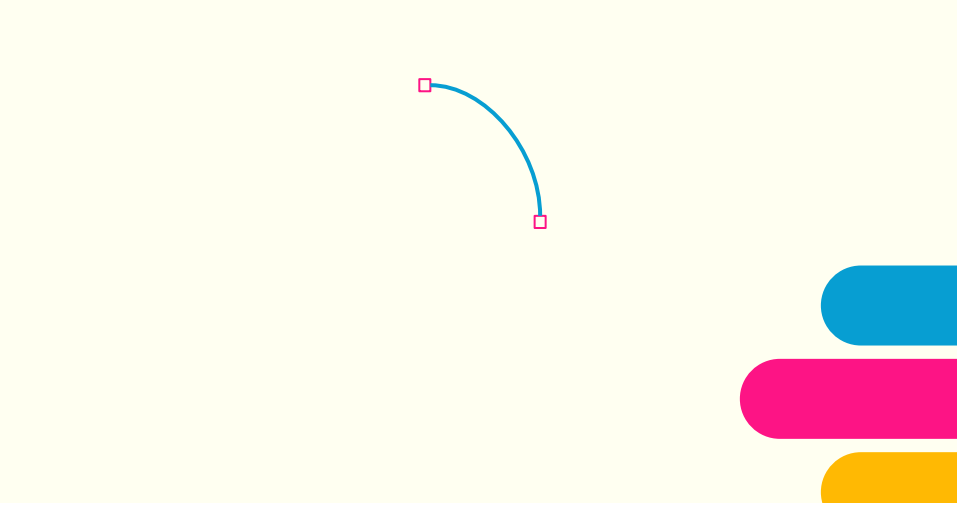
Maryland Census Data

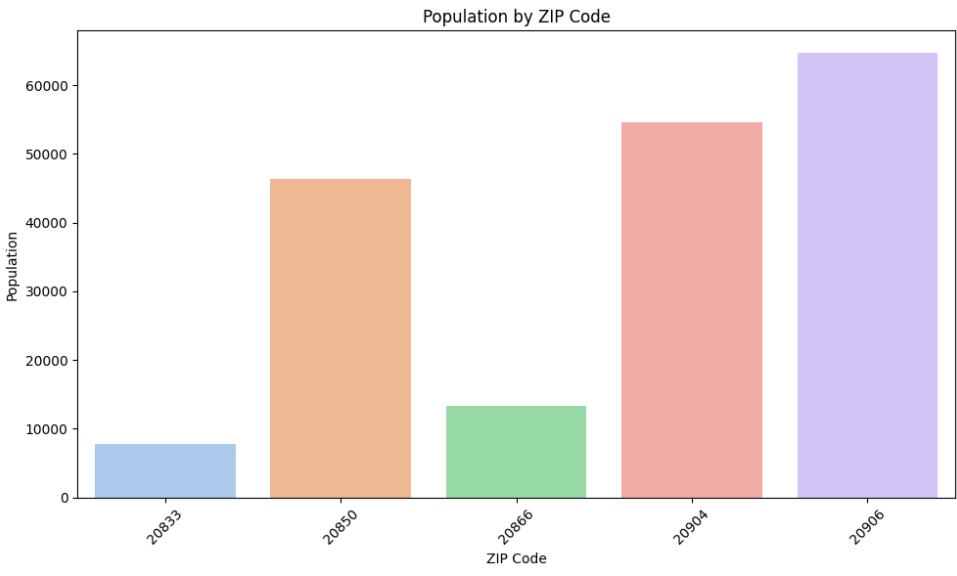
- Variables are population characteristics by ZIP code
- Source: Maryland.gov

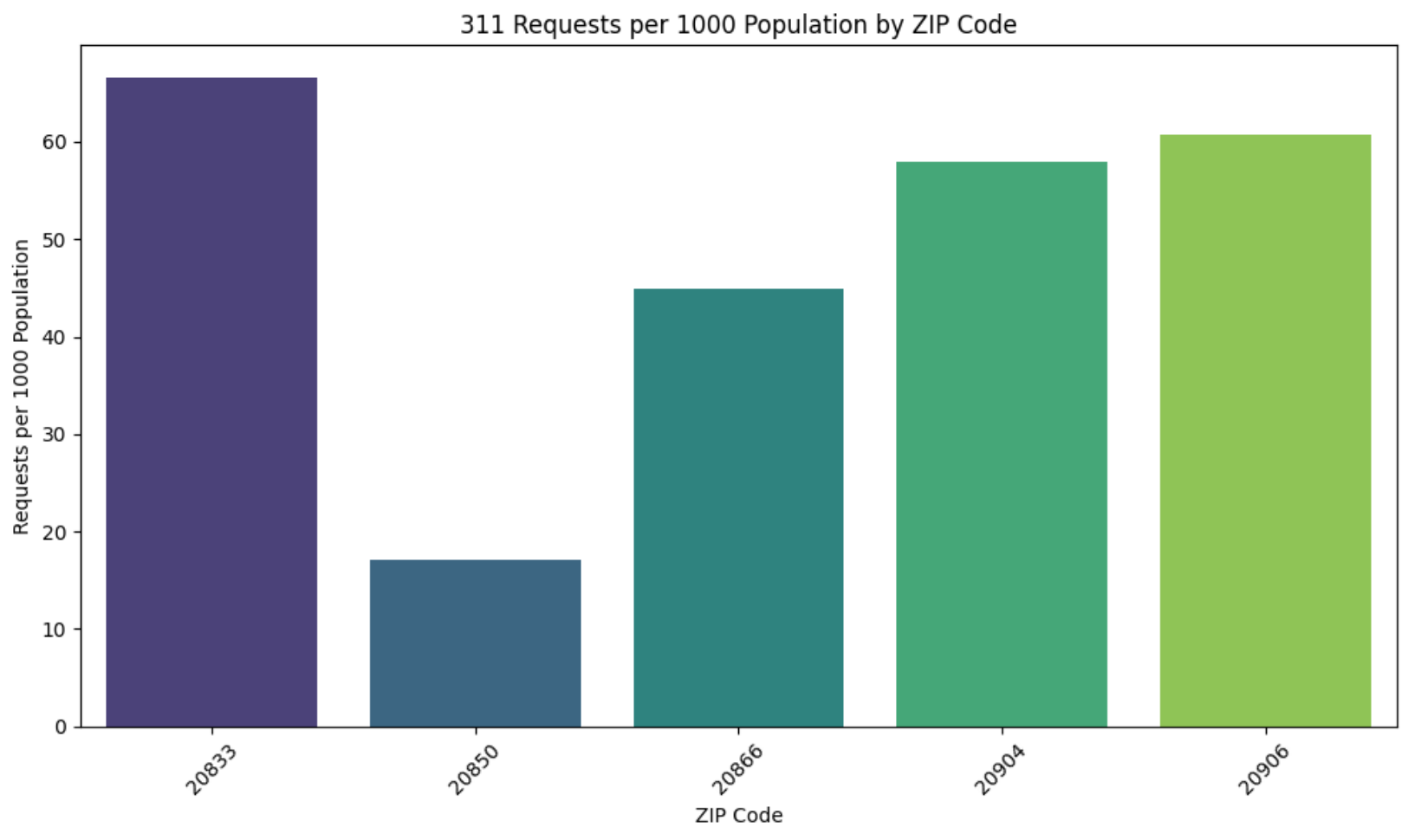
Ingestion

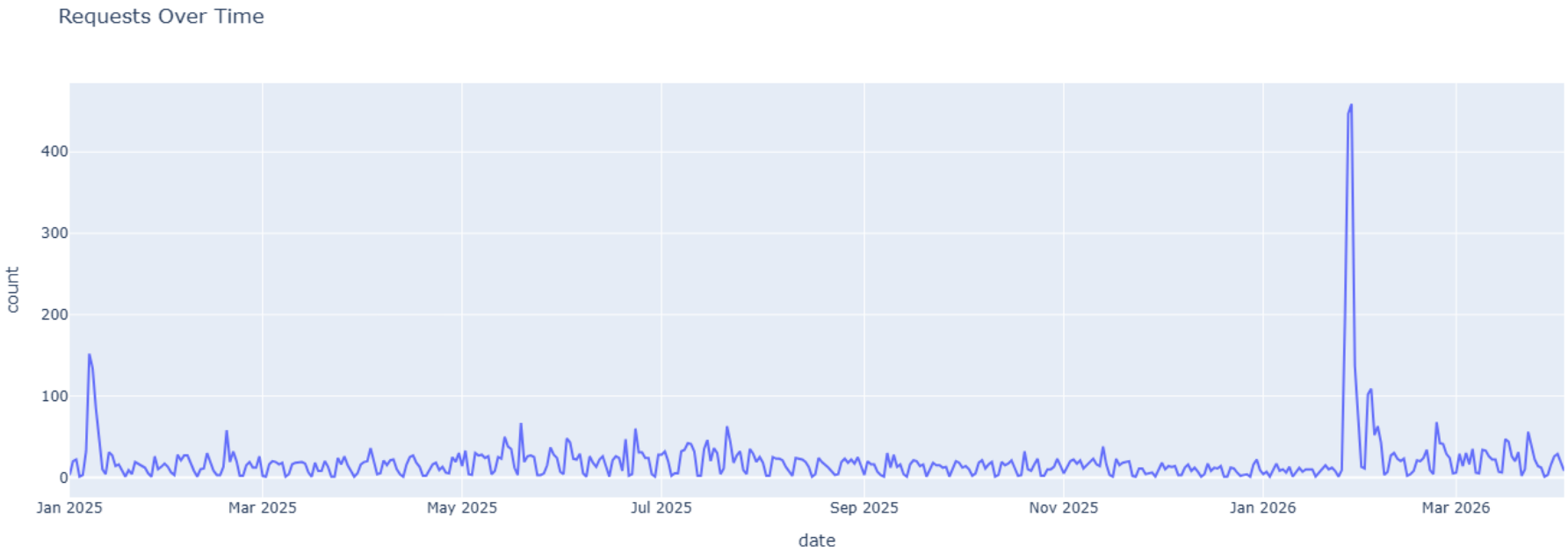
- SQL for selecting service areas and zip codes
- Date formats for MC311 data changed to match Weather data
- “response_time” variable created
- “avg_temp” variable created
- Merged MC311 and Weather
- Filtered/selected to chosen 5 zip codes for Zip Code data
- Renamed variable names for readability for all

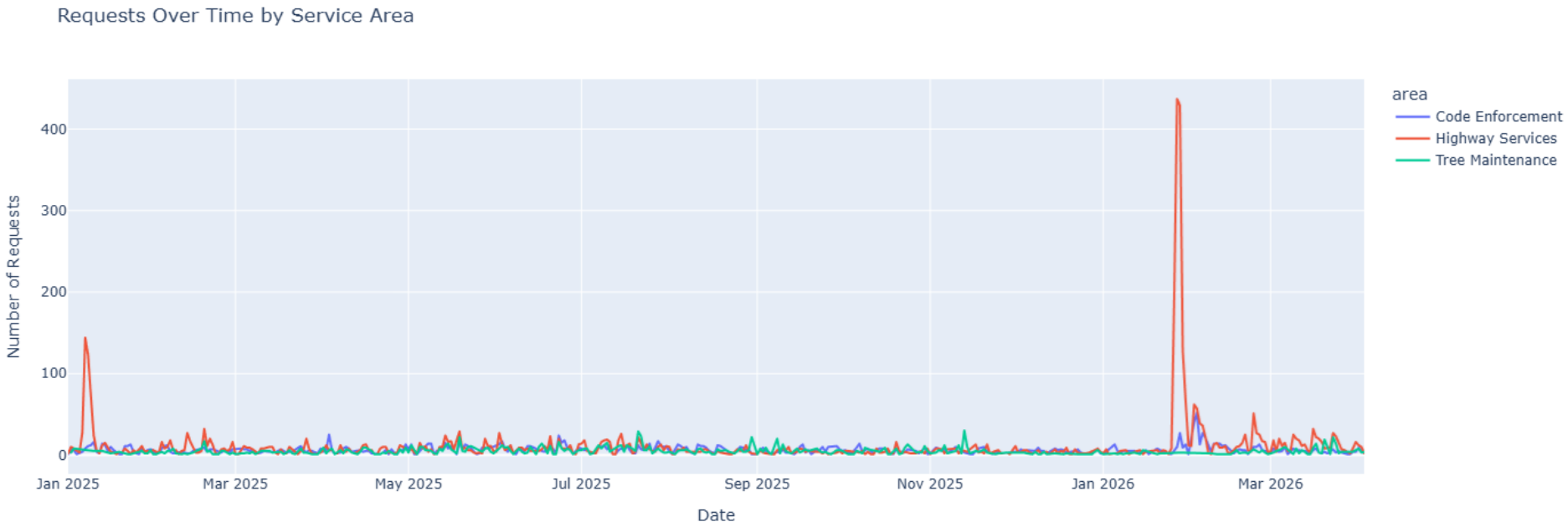

Requests by ZIP Code
count
4000
3500
3000
2500
2000
1500
1000
500
0
20833
20850
20866
20904
20906
zip_code

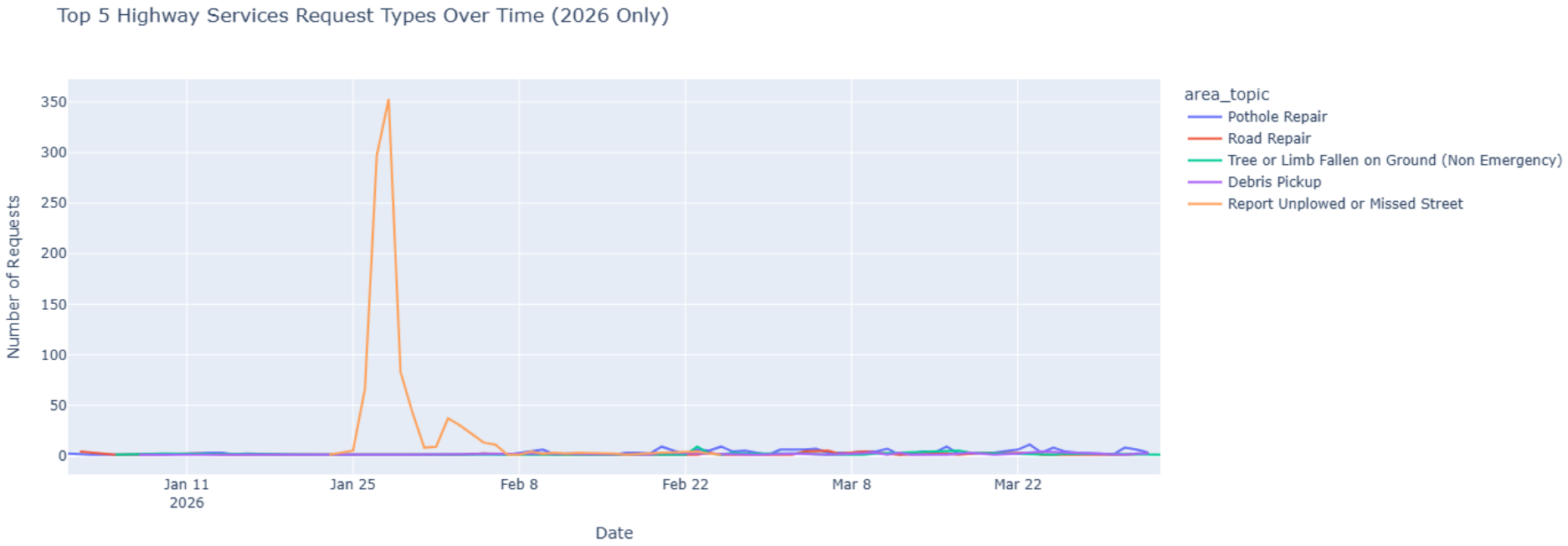


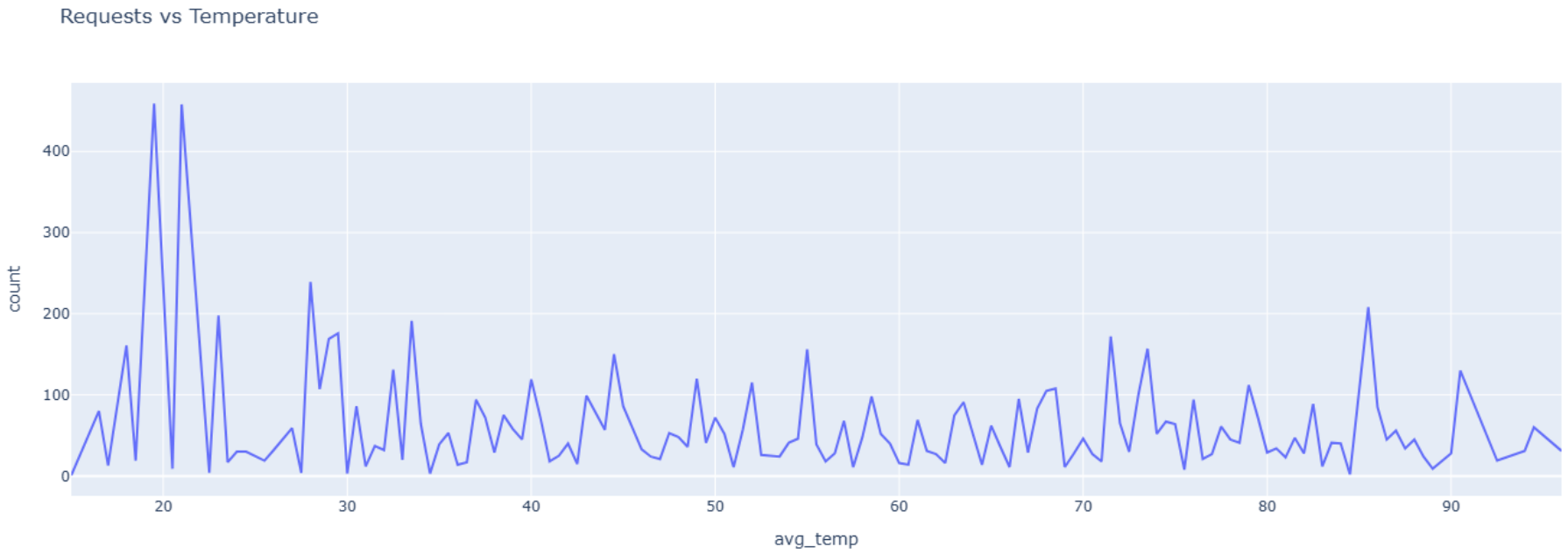

Population by ZIP Code
Population
60000
50000
40000
30000
20000
10000
0
20833
20850
20866
20904
20906
ZIP Code


311 Requests per 1000 Population by ZIP Code
Requests per 1000 Population
0
10
20
30
40
50
60
20833
20850
20866
20904
20906
ZIP Code

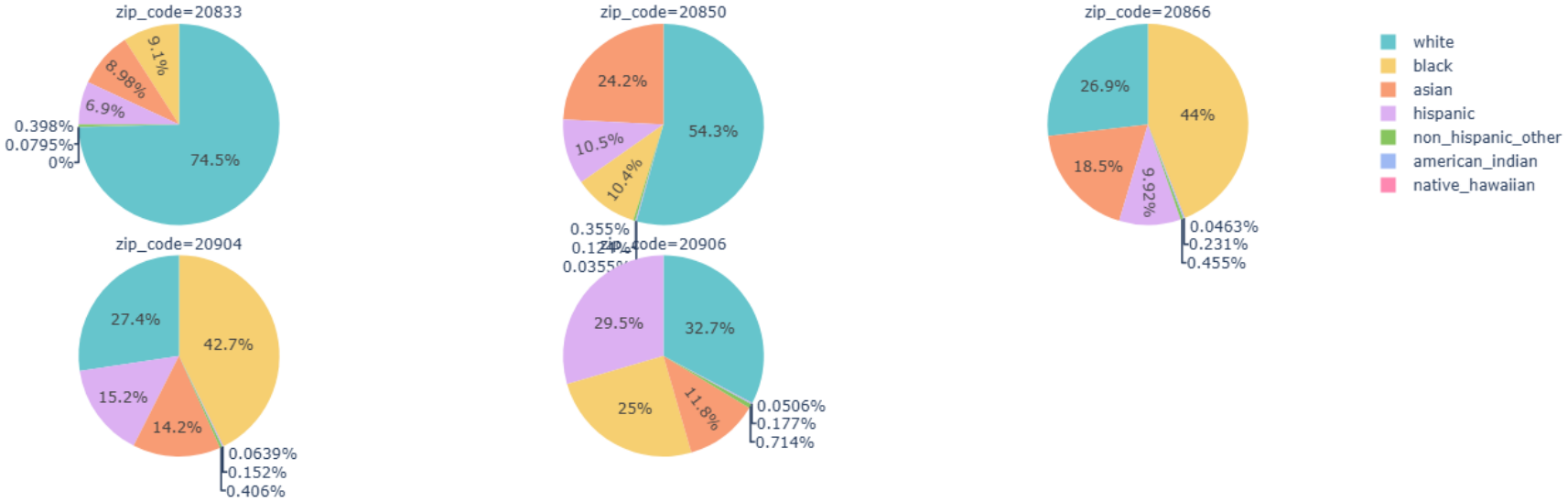

Requests Over Time
count
400
300
200
100
0
Jan 2025
Mar 2025
May 2025
Jul 2025
Sep 2025
Nov 2025
Jan 2026
Mar 2026
date


Requests Over Time by Service Area
Number of Requests
400
300
200
100
0
Jan 2025
Mar 2025
May 2025
Jul 2025
Sep 2025
Nov 2025
Jan 2026
Mar 2026
Date
area
Code Enforcement
Highway Services
Tree Maintenance


Top 5 Highway Services Request Types Over Time (2026 Only)
Number of Requests
350
300
250
200
150
100
50
0
Jan 11
2026
Jan 25
Feb 8
Feb 22
Mar 8
Mar 22
Date
area_topic
Pothole Repair
Road Repair
Tree or Limb Fallen on Ground (Non Emergency)
Debris Pickup
Report Unplowed or Missed Street


Requests vs Temperature
count
400
300
200
100
0
20
30
40
50
60
70
80
90
avg_temp

Racial Composition by ZIP Code (Percentage)



Next Steps

- **Examine relationships between variables with correlation analysis**
  - Strength and direction of relationships weather vs. demand, etc.
- **Analyze weather impact with linear regression**
- **Evaluate socioeconomic effects using multiple regression**
  - Housing characteristics, racial compositions and age relation to demand across ZIP codes
- **Model SLA outcomes using logistic regression**
  - Examine factors affecting whether requests meet SLA (Yes/No)